

Saransh Agrawal

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🌐 [Website](#)

🐙 [GitHub](#)

Education

Texas A&M University

Master of Science in Data Science: GPA: 3.7/4.0

Aug 2023 – May 2025

College Station, TX

Manipal Institute of Technology

Bachelor of Technology in Control Systems Engineering

July 2017 – Aug 2021

Manipal, IN

Technical Skills

Languages: Python, Go, C++, SQL.

ML/AI Frameworks: PyTorch, Transformers, TensorFlow, Scikit-Learn, HuggingFace, vLLM.

MLOps & Infrastructure: Docker, Kubernetes, Apache Kafka, AWS (EC2, SageMaker, S3), SLURM, Jenkins CI/CD, Git.

Data & Storage: PostgreSQL, MongoDB, Elasticsearch, Apache Spark, Pandas, NumPy.

Areas: Applied ML, Large Language Models, Natural Language Processing, Reinforcement Learning, Distributed Systems.

Publications

- [1] **Saransh Agrawal** and Kuan-Hao Huang. Selective Amnesia – Constrained Unlearning for Large Language Models via Knowledge Isolation. In *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025) at ACL*.
- [2] Ren-Wei Liang, Chin-Ting Hsu, Chan-Hung Yu, **Saransh Agrawal**, Shih-Cheng Huang, Shang-Tse Chen, Kuan-Hao Huang, and Shao-Hua Sun. Adaptive Helpfulness-Harmlessness Alignment with Preference Vectors. *arXiv:2504.20106* (under review)

Work Experience

Research Assistant

Aug 2024 – May 2025

FLAIR Lab, Texas A&M University

College Station, TX

- Developed a novel Preference Vector framework for **LLM alignment**, for dynamic and controllable balancing of behaviors at inference. Applied **DPO**, fine-tuning **Llama3 (8B, 3B)** and **Mistral-7B**, achieving superior performance over Safe-RLHF baseline in collaboration with global team from NTU.
- Pioneered a two-stage methodology for targeted **unlearning in LLMs**, effectively removing sensitive information while preserving general model capabilities. **2nd place** in SemEval-2025 Task 4 (presented at **ACL '25**).
- Utilized causal mediation analysis to pinpoint knowledge localization within **OLMo** models and designed a layer-specific constrained optimization strategy for precise knowledge removal.
- Architected and built a comprehensive framework for **benchmarking LLMs** on the MMLU dataset across diverse GPU configurations on **HPC** cluster using **SLURM**.
- Developed an ML optimization framework that reduced memory consumption by 75% for **long-context multimodal reasoning** by implementing Streaming LLM and Cache Merging techniques.

Software Developer, ML Team

July 2021 – July 2023

Viewzen Labs

Bangalore, India

- Led predictive modeling initiative to identify at-risk women from a **highly imbalanced** dataset with 300+ features.
- Architected** and spearheaded the development of a **scalable**, distributed AutoML platform (**MLaaS**) designed for **high-throughput**, processing over 800K requests monthly for 12+ enterprise clients.
- Designed a **robust** microservices architecture using **REST APIs** and **Kafka** to handle **millions** of data points daily, ensuring system reliability and maintainability.
- Built Python library implementing AutoML logic using Hyperopt, Scikit-Learn, reducing ML development time by 80%.
- Created RESTful Python microservices with Docker containerization. Mentored 2 junior developers on API design.
- Improved processing speeds by 20% with **C++** optimization and **concurrency** for **high-volume data transformation**.

Machine Learning Engineer Intern

Feb 2021 – June 2021

Centre for Digital Financial Inclusion

Delhi, India

- Transformed **raw audio signal** to its **MFCC** components using custom TensorFlow layers, for **Speech-to-text**.
- Implemented RNN, GRU and LSTM architectures for text prediction at character, word and phoneme level.
- Utilized **DeepSpeech** transfer learning on Indian Accent dataset achieving **0.10 CER** and **0.17 WER** with KenLM language model. Added **T5 Transformer** to generate **SQL queries** and subsequent visualizations.

Projects

- **Satellite-based Crop Monitoring System (SCMS)** — *Python, FastAPI, Docker, React.js, MongoDB*
 - Led 4-person team building data ingestion, transformation, and serving pipelines for satellite imagery processing.
 - Integrated PlanetScope satellite API for automated imagery acquisition with cloud filtering and AOI-specific preprocessing.
 - Deployed **LSTM-based time-series forecasting** models to predict outcomes based on sequential satellite data, achieving an R^2 of **0.81**. Presented at **ASA Conference 2024**.
- **Graph-Based Ranking & Semantic Search Engine** — *Go, Python, Apache Arrow, Sentence-Transformers*
 - Engineered an end-to-end **search** system for 70k+ ACL Anthology papers, improving relevant content discovery.
 - Constructed a large-scale citation graph and implemented the PageRank graph algorithm from scratch to compute paper authority scores for **targeted content discovery**.
 - Built hybrid search system combining graph-based metrics and **sentence embeddings** to model complex relationships.